

Eric Dong

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EDUCATION

University of Louisville | B.A. in Computer Science May 2027

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services Jul 2026
Credential ID: afd04de2-aaf7-45c6-86fb-3a0700c53689

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, HTML, Bash
Development stack: Flask, React, MySQL, Nginx, REST APIs, Git, GitHub, Storybook
Cloud & Infrastructure: AWS (EC2, S3, IAM, Lambda, RDS, VPC, CloudWatch), Linux

SELECTED PROJECTS

Google A2UI Composer Component Automation

- Building developer tooling to automate conversion of reusable React and Storybook components into A2UI-compatible JSON catalogs for AI-driven UI generation.
- Developing base component catalogs for ShadCN and Material Design 3, with plans to extend the catalog to higher-level finance and healthcare components.
- Integrating A2UI Composer with the Gemini API to support searchable component discovery and interactive UI generation.

AWS Web Application – Car and Driver Database

- Developed and deployed a full-stack web application on AWS EC2 by integrating Flask, MySQL, and Nginx to provide cloud-hosted vehicle and driver management.
- Configured Linux server networking, security groups, and web server deployment to securely host a production-ready application.
- Troubleshoot deployment, networking, and database issues to ensure stable cloud application performance.

Arduino Water Filtration System

- Designed and programmed a C++ microcontroller-based water filtration system that achieved approximately **95% detection accuracy** during testing.
- Simulated real-world turbidity detection by integrating sensor logic into an embedded system prototype.
- Improved filtration decision-making by developing control logic for automated sensor responses.

Python Vector Calculator

- Developed a Python application that calculated geographic distances using vector estimation and the Haversine formula.
- Parsed Google Earth coordinate data to automate accurate distance calculations between locations.

WORK EXPERIENCE

University of Louisville | *Student Researcher* Jul 2022 – Aug 2023

- Selected as **1 of 13 students** for the UofL Cancer & Health Disparity Summer Bridge Program.
- Optimized glioblastoma tissue decellularization procedures using scanning electron microscopy, confocal microscopy, and immunostaining.
- Conducted laboratory research that supported ongoing biomedical engineering investigations into brain cancer.

University of Louisville | *Student Researcher* Jul 2021 – Aug 2021

- Selected as **1 of 25 students** for the competitive Louisville Science Pathway Program.
- Developed and maintained a mutant strain database for *Burkholderia pseudomallei*, improving laboratory organization and accessibility.
- Performed bacterial conjugation and transposon mutagenesis to generate mutant strains for microbiology research.
- Applied antibiotic selection techniques to validate experimental bacterial strains.

ORGANIZATIONS

River Lotus Lion Dance – Volunteer Jan 2023 – Present
Beta Theta Pi – Member Jan 2024 – Present